SE(3)-WAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

Figure placeholder: concept figure, design pending

Planned content: one Dynamic-14 frame overlaid with its real training labels; no generated imagery. The target at capture (solid) and when the commands take effect (translucent); the target's SE(3) poses at the 8 forecast horizons (axes); sparse 3D flow on the 14×14 patch points (arrows, near zero on the background). Task, layout, and styling to be decided.

Figure 1. **Concept figure (placeholder).** The caption will be written with the final figure.

Abstract

Manipulating moving objects requires actions that fit the scene when they execute, not when the image was captured, yet every millisecond spent inferring the scene's motion lets the target move further. World–action models forecast future images or latents at uniform fidelity, spending time on appearance while leaving the target's metric pose implicit; target-centric predictors are fast but ignore the surroundings that constrain or drive the motion. We observe that the parts of a dynamic scene need different precision: the target's pose decides contact to within millimeters, the bodies around it matter only through the clearance they leave and the motion they impart to the target, and the remaining background does not matter. We present SE(3)-WAM, a world–action model built on this role-aware 4D structure. It encodes the target's motion as a metric SE(3) pose history, so one RGB-D frame suffices as input. It learns to forecast the target's future SE(3) trajectory precisely and the whole scene coarsely as sparse 3D flow, with forecasts used only in training. Together, these choices let SE(3)-WAM act in [XX] ms on one RTX 5090, [N]× faster than video world–action baselines. We also introduce Dynamic-14, an open-source benchmark of 14 simulated dynamic manipulation tasks in which the scene keeps moving throughout each policy's inference. SE(3)-WAM achieves [XX.X]% success on Dynamic-14, [XX.X] points above the strongest baseline, and [XX]% across six real-robot tasks with diverse target motion.

1. Introduction

Learning-based manipulation has advanced rapidly, from diffusion policies to vision–language–action (VLA) and world–action models (WAMs) [5, 6, 12, 21, 43, 49, 63], yet most of this progress assumes that objects stay still unless the robot moves them. In factories, kitchens, and human–robot handovers, objects keep moving while the robot perceives, computes, and acts. In such scenes, an action must fit the scene at the moment it executes, not the moment its observation was captured, so a policy has to know how the scene moves. That knowledge must itself be inferred, and inference takes time during which the scene keeps moving: the longer a policy computes to anticipate the motion, the further ahead it must anticipate.

Recent dynamic policies add a memory of past motion, typically image-space cues such as optical flow [14, 28], and a forecast of the future [28, 39, 56]. Video-based WAMs forecast whole future frames or latents at uniform fidelity, spending much of their computation on visual detail [43, 49]. Compact future codes [29] reduce this cost and metric 3D scene dynamics [54] add geometry, but both still treat the whole scene uniformly. In each case, the quantity that decides contact, the target's metric pose at execution time, stays implicit. Target-centric predictors and dynamic VLAs with object-level forecasts sit at the other end [14, 18]: they are fast but anticipate only the target, leaving out the surroundings that can block or redirect its motion. Neither approach allocates precision according to what contact requires from each part of the scene. This raises a central question: what must a policy know about a moving scene, and how precisely, to make contact while keeping inference latency low?

Our key observation is that a dynamic policy needs to know very little precisely (Fig. 1). Contact demands millimeter precision in the target's pose at execution time; the surroundings matter only through the clearance they leave and the motion they impart to the target, so they need only be known coarsely. Building on this observation, we introduce SE(3)-WAM, a world–action model that learns a role-aware metric 4D representation, precise where contact demands it (Fig. 2). To supply motion information without encod-

ing past image frames, SE(3)-WAM pairs a single RGB-D frame with a short history of the target's metric SE(3) poses. A world stream learns the representation by forecasting the target's future SE(3) trajectory and the motion of the whole scene as a sparse 3D flow. An action stream reads the world stream's scene and summary tokens to generate action chunks, with explicit forecasting confined to training [53]. SE(3)-WAM thus combines precise anticipation of target motion with low inference latency, limiting how much the scene changes between observation and action.

We build Dynamic-14, an open-source benchmark of 14 simulated dynamic manipulation tasks in which the scene keeps moving throughout each policy's inference, so every method pays for its own latency. SE(3)-WAM achieves [XX.X]% success on Dynamic-14, [XX.X] points above the strongest baseline, at a p95 latency of [XX] ms, [N]× lower than that of video world–action baselines. It also leads on 24 DOMINO tasks [14] and succeeds in [XX]% of trials on six real-robot tasks, against [XX]% for the strongest baseline. Controlled ablations test the contributions of target-motion history and future supervision, and compare role-aware with uniform-fidelity forecasts to examine the allocation of precision.

Our contributions are threefold:

- We show that a dynamic policy needs millimeter precision only in the target's pose at execution time, and instantiate this principle as role-aware 4D modeling: a precise metric SE(3) trajectory for the target and a coarse sparse 3D flow for the whole scene.
- We develop SE(3)-WAM, a world–action model that combines metric target-pose histories with role-aware predictive supervision for low-latency action generation.
- We introduce Dynamic-14, an open-source dynamic manipulation benchmark that charges each policy its own inference latency, and demonstrate gains over strong baselines on Dynamic-14, 24 DOMINO tasks, and six real-robot tasks.

2. Related Work

Dynamic manipulation. Compensating for the time between sensing and acting is a classical problem, from control with dead time [36] to catching systems that track the target with Kalman-type filters and plan an interception [1–3, 31]; such systems are precise but engineered for each object and task. Learned policies have recently been brought to moving scenes [47]. PUMA reads historical optical flow and is supervised with future object features [14], and DynamicWAM conditions a WAM on flow and kinematic descriptors and executes it with real-time chunking [28]. Both infer past motion from image-space cues, whereas SE(3)-WAM takes the target's metric SE(3) pose history as input and observes only the current RGB-D frame. Other dynamic policies forecast salient tokens [39], future images [57], compact future codes [29], or 3D scene dynamics [54] and condition their actions on these forecasts at test time. DynaWM adapts a base VLA's action chunks for dynamic manipulation [10], and DynamicManip synthesizes dynamic demonstrations from a static one [23]. Execution schemes reduce or hide latency by overlapping inference with execution [34, 47], keeping consecutive chunks consistent [7, 26], or conditioning on the execution time [40]. They decide when actions are computed and applied, not what the policy represents. DOMINO [14] and PhysMani-Bench [54] provide diverse dynamic-manipulation tasks, but their released default evaluation pipelines advance the simulation only after a policy has returned its prediction. Dynamic-14 advances the scene during inference and charges each method its measured latency.

Predictive supervision in WAMs. World–action models (WAMs) generate future video and actions in one network [22, 43, 49, 61], building on video prediction as a policy objective [41, 46]. Because generating video is costly, recent work changes what is forecast, such as dynamic regions, depth, and semantic features [56], flow and geometry alongside RGB [52], object masks or slots [27, 50], or compact latents [38, 60], and asks whether the forecast is needed at test time. Fast-WAM retains most of the benefit of video modeling without test-time imagination [53], and PUMA and SG-WAM drop their predictors after training [14, 58]. SE(3)-WAM keeps this training-only use but forecasts primarily a metric 4D state rather than pixels, features, or latents, so the target's future pose is supervised explicitly instead of being left implicit.

Object-centric and geometric motion representations. Metric 3D inputs make policies more precise and sample-efficient [33, 55]. Motion has been represented by 2D point tracks [4, 45, 48], 3D flow [16, 19, 51, 54], and object keypoints or SE(3) trajectories [18, 30, 37], while rigid-body decompositions have long been used to learn scene dynamics [8, 9]. These works describe the scene with one kind of representation, such as dense flow or tracks everywhere, or a pose for each object. Target-centric predictors, including dynamic VLAs with object-level forecasts, model only the target's future [14, 18] and leave out the bodies that block or push it. DOCIR separates the robot, the target, and obstacles, but uses these distinctions only as input encodings [13]. SE(3)-WAM assigns precision by role: a coarse sparse 3D flow describes the whole scene, while the target, on which contact depends to within millimeters, is additionally described by its SE(3) history and forecast.

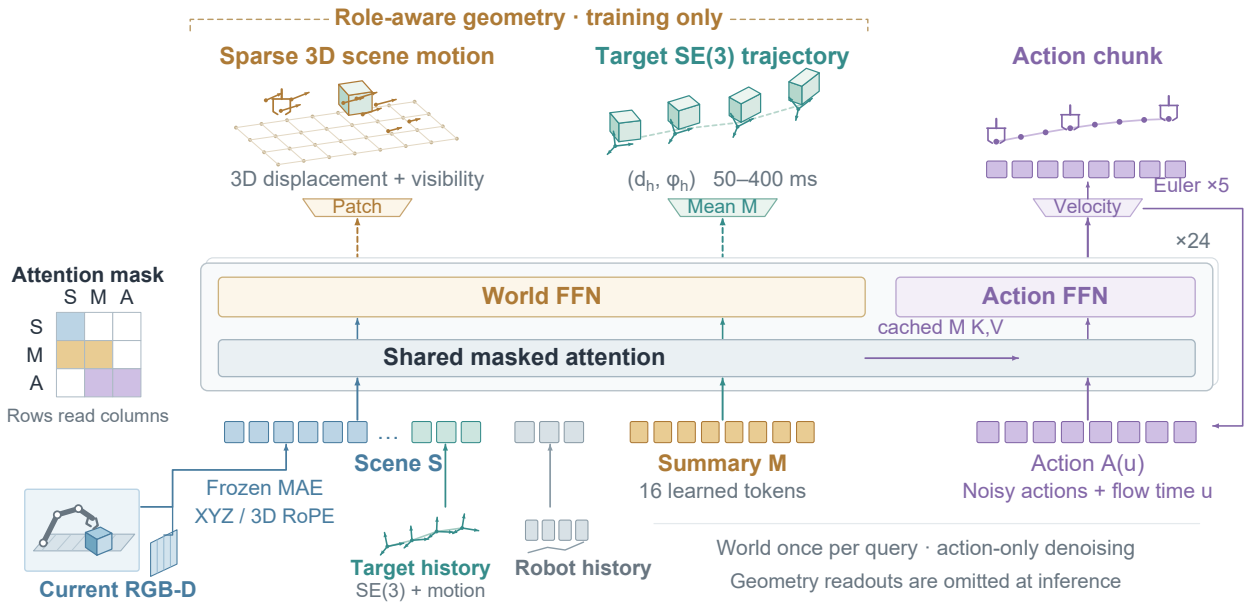


Figure 2. **SE(3)-WAM architecture.** Read from bottom to top. Current RGB-D and target/robot histories feed the world stream. World and action streams share attention and use separate FFNs. Training-only heads (dashed) predict sparse whole-scene 3D displacement and visibility from patch states, and target SE(3) trajectories from the mean summary. Actions read scene and summary tokens through the structured mask. At inference, scene and summary keys and values are computed once per query and reused across 5 Euler steps; forecast heads are skipped. Geometry drawings are schematic.

3. SE(3)-WAM

We formulate dynamic manipulation as generating time-indexed command chunks (Sec. 3.1) from a representation supervised, at the times these commands take effect, with the target's SE(3) trajectory and the scene's sparse 3D flow (Fig. 2). We describe the metric inputs (Sec. 3.2), this supervision (Sec. 3.3), and how an action stream reads the representation to generate the chunk (Sec. 3.4).

3.1. Problem Formulation

We study manipulation in scenes that keep moving while the policy computes. At query t, a policy π learned from demonstrations reads an RGB image I_t and a depth map D_t captured at time τ_t, together with histories $\mathcal{T}_t$ of the target pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$ and $\mathcal{R}_t$ of the robot state, all in the robot base frame, and outputs a chunk of H commands

$$(\mathbf{a}_{t,0}, \ldots, \mathbf{a}_{t,H-1}) \sim \pi\big(\cdot \mid I_t, D_t, \mathcal{T}_t, \mathcal{R}_t\big), \qquad (1)$$

where command j is meant for time $\tau_t + j\Delta_a$, with control period Δ_a.

Each task specifies a *target*, whose pose the robot must match within a contact or containment tolerance (Table S3). The surroundings that matter form the *context*: obstacles, which the robot must clear, and drivers, which move the target. The chunk becomes available only after inference, so each executed command must account for the target and the context at its execution time rather than at τ_t.

3.2. Metric Scene and Motion Inputs

SE(3)-WAM encodes the target's motion precisely and its context through the current geometry: the current RGB-D frame gives the metric layout of the scene, and a short history of target poses gives the motion that one frame does not show, which also reflects how drivers have moved the target.

Current scene. SE(3)-WAM anchors each patch token at a metric 3D point, so that the observation lives in the base frame in which the forecasts and the actions are expressed. A frozen MAE ViT-B/16 [17] turns the RGB image into 196 patch tokens on a 14×14 grid, and the depth map back-projects the center of patch i to a point $\mathbf{x}_i \in \mathbb{R}^3$ in the base frame. A linear embedding of $\mathbf{x}_i$ is added to the patch token, and $\mathbf{x}_i$ drives 3D rotary position embeddings of the patch queries and keys, so that attention between patches depends on their relative 3D positions. The points $\mathbf{x}_i$ are also the source points of the scene flow (Sec. 3.3).

Target motion. Instead of past frames, from which motion must be recovered by comparing images, SE(3)-WAM receives the target's recent motion as a metric SE(3) state. The history spans the last 350 ms in 8 samples at 20 Hz; its 8 tokens hold the current target pose and the seven motions

between consecutive samples, each with its translation, relative rotation, time interval, and finite-difference velocity and acceleration in physical units (Appendix A.2). The velocity and acceleration that decide where the target will be thus enter the model directly, at the cost of a few low-dimensional tokens; Sec. 4.3 removes this history to measure its effect. In simulation, the poses come from the simulator; on the robot, a causal tracker estimates them from the RGB-D stream (Appendix D).

World stream. The patch tokens, the target tokens, and 8 robot tokens with the end-effector states at the times of the target history form the 212 scene tokens. Together with 16 learnable summary tokens that aggregate them, they make up the world stream.

3.3. Role-Aware 4D Predictive Supervision

SE(3)-WAM supervises its representation with two geometric forecasts of different precision: the motion of the whole scene as a sparse 3D flow on the patch grid, and, because the target's pose decides contact, the full SE(3) trajectory of the target. Both are expressed in the base frame as changes from the current state at the times $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50\,\text{ms}$ for $h = 1, \dots, 8$, from 50 ms to 400 ms after capture; together with the current state, these horizons cover every time at which a command of the chunk takes effect (Appendix B.3).

Scene flow. At the resolution of the patch grid, the flow covers every surface that the patches see: the robot, the target, and the context, whose clearance and motion the commands must account for (Sec. 3.1). For each of the 196 points $\mathbf{x}_i$ and each horizon h, a linear head on the final state of patch token i predicts the point's base-frame displacement $\hat{F}_{ih} \in \mathbb{R}^3$ and a visibility logit $\hat{o}_{ih}$, trained with

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h)\in\mathcal{V}_F} \Big[\tfrac{1}{3} \big\| (\hat{F}_{ih} - F_{ih})/s \big\|^2 + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \Big], \tag{2}$$

where F_{ih} is the displacement of the surface point at $\mathbf{x}_i$, o_{ih} its visibility, s a fixed scale fitted on the training labels, and $\mathcal{V}_F$ the set of valid labels.

Target trajectory. The flow samples the target only at the few patches it covers, too sparse to determine its rotation, so a dedicated linear head on the mean of the final summary states forecasts its rigid motion directly: for each horizon h, the target's displacement $\hat{\mathbf{d}}_h$ and a rotation increment $\hat{\boldsymbol{\phi}}_h$, trained with

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h\in\mathcal{V}} \Big[\|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \Big], \tag{3}$$

where $\mathbf{d}_h = \mathbf{p}(\tau_t{+}\delta_h) - \mathbf{p}(\tau_t)$ and $R_h = R(\tau_t{+}\delta_h)$ are the true displacement and orientation, $\hat{R}_h = \mathrm{Exp}(\hat{\boldsymbol{\phi}}_h^{\wedge}) R(\tau_t)$, $d_{\mathrm{SO}(3)}$ is the geodesic distance, and $\mathcal{V}$ is the set of horizons with valid labels. The rotation term matters for contact: a rotation error θ can displace a contact point at distance r from the target's origin by up to $2r\sin(\theta/2)$. For example, a 5° error at $r = 10\,\text{cm}$ can shift the contact point by nearly 9 mm.

From supervision to actions. Scene-flow supervision acts on the patch states, and target-trajectory supervision acts on the summary states. The action stream reads both scene and summary tokens (Sec. 3.4), so it has direct access to the representations shaped by both geometric objectives.

3.4. World–Action Learning and Inference

Action stream. The action stream generates the chunk by conditional flow matching [24]. It holds $H = 8$ tokens, each embedding one noisy command: an end-effector translation and rotation increment and a gripper command (Appendix A.2). Writing S, M, and A for the scene, summary, and action tokens, the attention mask allows $S \to S$, $M \to \{S, M\}$, and $A \to \{S, M, A\}$, where $X \to \mathcal{Y}$ means that tokens in X attend to tokens in $\mathcal{Y}$. The world stream therefore never reads the actions. In every layer, the two streams use the same attention projections and separate feed-forward networks; world tokens use layer normalization, and action tokens use adaptive layer normalization conditioned on the flow time.

Joint learning. For a chunk $\mathbf{a}_1$ normalized with training statistics, noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and flow time $u \sim \mathcal{U}[0, 1]$, the action stream v_ψ regresses the flow-matching velocity

$$\mathcal{L}_A = \mathbb{E}\,\mathrm{MSE}\big(v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t),\, \mathbf{a}_1 - \mathbf{a}_0\big), \tag{4}$$

where $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, $\mathcal{S}_t$ denotes the inputs in Eq. (1), and the MSE averages over valid commands and action dimensions. The world and action streams are trained jointly with

$$\mathcal{L} = \mathcal{L}_A + \lambda_P \mathcal{L}_P + \lambda_F \mathcal{L}_F + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}, \tag{5}$$

where $\mathcal{L}_{\text{aux}}$ sums four auxiliary appearance and reconstruction losses (Appendix A.2). We set $\lambda_P = \lambda_F = \lambda_{\text{aux}} = \lambda_R = 0.1$.

Inference. At deployment, SE(3)-WAM skips the forecast heads, as recent world–action models do [22, 53, 56]. Because world tokens attend neither to the actions nor to the flow time, the world stream runs once per query, and the action stream integrates 5 Euler steps on the cached keys and values of the scene and summary tokens. Because commands are indexed by time from capture, the policy needs

no latency input: the executor hands over each chunk at the first control-period boundary at or after inference completes and starts from the command indexed to that boundary (Appendix B.3). On the robot, the latency of a query also includes tracking the target in the newest frame.

4. Experiments

We evaluate how SE(3)-WAM performs when the scene keeps moving during inference (Sec. 4.2), what each of its components contributes (Sec. 4.3), and how it transfers to a real robot (Sec. 4.4).

4.1. Setup

Dynamic-14. Dynamic-14 is a simulated benchmark in which observation, scene motion, and action execution share one physical clock, so the scene keeps moving while a policy computes and every method pays for its own latency. It contains fourteen tasks in MuJoCo/robosuite [42, 62] in five families: grasping a moving target (harmonic, conveyor, translating, rotating, and occluded motion), placing into a moving receptacle, passing a wand along a rotating wire, capturing rolling or sliding objects, and grasping after an impact (Fig. 3; Appendix B). Experts with privileged simulator state provide 800 demonstrations per task. We select configurations on development scenes and report all Dynamic-14 results on 1,400 held-out scenes (100 per task, new seeds).

Evaluation. During each query, physics advances by the method's own inference latency, drawn from its latency distribution measured at batch size 1 on one RTX 5090, the GPU of our robot, independently of the hardware that runs the rollouts (Appendix B.3). All methods share one executor that indexes each chunk by time from capture: a query starts at a control-period boundary, and its chunk takes over at the first boundary at or after the query ends (Appendix B.3). Besides this native-latency setting, we add 150 ms of latency to every method and pause physics during inference; Appendix C.2 adds equal-latency and original-executor controls. We report the success rate (SR), averaged over the fourteen tasks with equal weight and over training seeds, and the p95 latency, the measured time from an observation to its first command (time-to-action). Paired 95% intervals come from a bootstrap over held-out scenes.

Baselines. We compare with reactive policies, dynamic VLAs, image- or latent-space WAMs, and a dynamic WAM (Table 1). Each baseline is trained on the same demonstrations with its official code, inputs, and recipe, including a rendered wrist camera and frame or flow histories where the recipe uses them (Appendix B.5). SE(3)-WAM uses one agent view with its depth and the target's SE(3) history, without past frames. To give baselines the same target-motion information, $\pi_{0.5}$ and DynamicWAM also run with our SE(3) history tokens (+SE(3)).

Table 1. **Main results.** Task-balanced SR on 1,400 held-out Dynamic-14 scenes: native latency, +150 ms, and paused physics. Three training seeds for SE(3)-WAM, one per baseline. p95: time-to-action on RTX 5090. Wrist and history denote extra inputs (Sec. 4.1). Blue/bold and light blue mark the best and second reported values per column; rankings remain provisional while results are pending.

Method	Wrist	History RGB	History Flow	History SE(3)	SR (%) ↑ Native	SR (%) ↑ +150 ms	SR (%) ↑ Paused	p95 (ms) ↓
ACT [59]					**42.0**	**24.1**	**40.9**	**10.7**
DP [12]		✓			0.0	0.0	33.1	1701.9
$\pi_{0.5}$ [5]					—	—	—	—
$\pi_{0.5}$+SE(3)	—			✓	—	—	—	—
DynamicVLA [47]		✓			—	—	—	—
PUMA [14]			✓		—	—	—	—
OpenWAM-α [43]					—	—	—	—
DynamicWAM [28]			✓		—	—	—	—
DynamicWAM+SE(3)	—		✓	✓	—	—	—	—
SE(3)-WAM				✓	—	—	—	—

Implementation. The transformer has 24 layers of width 1024 with 16 heads and 523.8M trainable parameters; the MAE ViT-B/16 encoder and the DINOv3 teacher stay frozen. On Dynamic-14, it trains from random initialization for 20 epochs with AdamW and the loss weights of Sec. 3.4 (Appendix A.3). On the real robot, it is fine-tuned per task from a 48-layer base pretrained on DROID [20] (Appendix A.4).

4.2. Simulation Results

Comparison. With its native latency, SE(3)-WAM reaches [XX.X]% SR on held-out Dynamic-14 scenes (Table 1), [X.X] points above the strongest baseline, [name], and [X.X] points above the strongest baseline that receives the same SE(3) history, [name]. Its gains are largest on [task types] (Appendix C.1). Its p95 latency of [XX] ms is [X]× lower than that of the image-space WAMs and close to that of [the reference reactive policy, to be fixed].

Latency and protocols. With 150 ms of added latency, SE(3)-WAM keeps [XX.X]% SR, whereas the reactive policies drop to [XX.X]% and the image-space WAMs to [XX.X]% on average, as Eq. S1 predicts: added latency lengthens the lead that multiplies every velocity error, so methods that do not anticipate the target's motion lose most. Fig. 4 shows the available measurements in the 0–200 ms sweep; pending points are listed in Appendix C.2. With

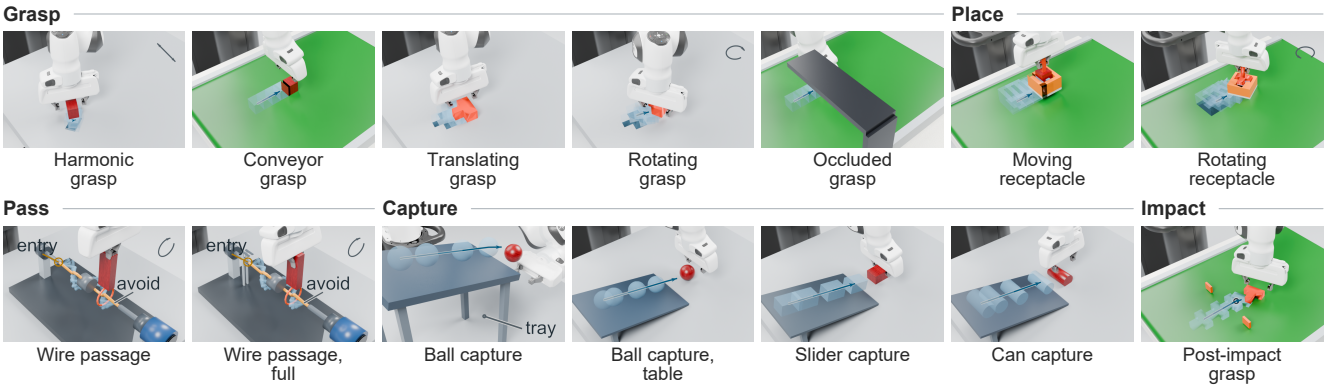


Figure 3. **Dynamic-14.** Fourteen tasks in five families, rendered from recorded simulator states with a common agent view. Four translucent blue poses fade from recent to earlier history; navy traces follow measured past motion. Gray corner arrows indicate rotation or oscillation qualitatively. A small ring marks the target center during recorded impact. Both Pass panels show passage: the wire is a target at entry and an obstacle along the path; Full additionally requires acquisition, exit and release. Occluded motion stays hidden. These native-expert examples illustrate task dynamics, not SE(3)-WAM outcomes; no future prediction is shown.

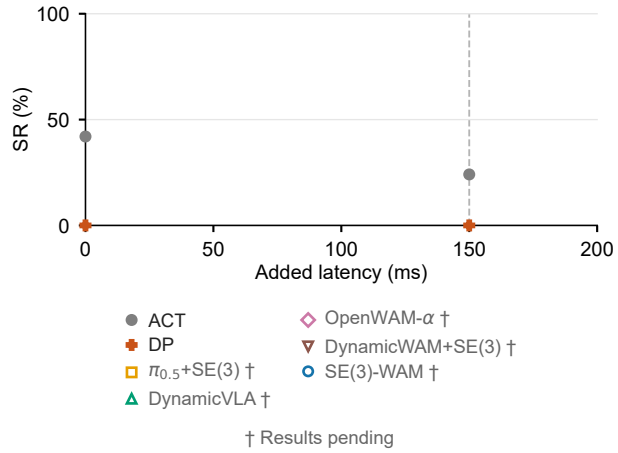


Figure 4. **Success under added latency.** Comparison on 1,400 held-out Dynamic-14 scenes. The legend retains all planned series, plus DP; † marks methods awaiting results. ACT and DP have measurements at 0/150 ms. Only adjacent measured points are connected. Dashed line: +150 ms; full sweep table: Table S7.

physics paused during inference, which ignores the cost of computation, SE(3)-WAM still leads [name] by [X.X] points, against [X.X] points with native latency. When every method is charged the p95 latency of SE(3)-WAM, its margin is [X.X] points, and running [the baselines that rely on their own asynchronous execution] with their original executors changes their SR by at most [X.X] points (Appendix C.2).

DOMINO. We also train SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of DOMINO [14] with its demonstrations and evaluate them with its native evaluator. SE(3)-WAM reaches [XX.X]% SR against [XX.X]% for [the strongest baseline] (Table 2; Appendix C.3). As on Dynamic-14, it reads the simulator pose of the target.

Table 2. **DOMINO results** on 24 single-arm clean-L1 tasks, each method trained on DOMINO's demonstrations (one training seed per method, 50 episodes per task and method). SR (%) is grouped by interaction type and averaged over tasks; MS is the manipulation score. Best per column will be bold on blue once measured. Protocol: Appendix C.3.

Method	Intercept	Track	All ↑	MS ↑
$\pi_{0.5}$	—	—	—	—
DynamicWAM	—	—	—	—
PUMA	—	—	—	—
SE(3)-WAM	—	—	—	—

Table 3. **Real-robot results.** Success rate (%) on the six tasks of Sec. 4.4, 20 trials per task and method; All pools the trials. p95 lat.: p95 latency, the end-to-end time-to-action (ms) on one RTX 5090, including tracking for SE(3)-WAM. DP: Diffusion Policy. Once measured, best per column will be bold on blue, including ties. Successes, intervals, and failures in Appendix D.

Method	T1	T2	T3	T4	T5	T6	All ↑	p95 (ms) ↓
$\pi_{0.5}$	—	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—	—
DP	—	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—	—

4.3. Ablations and Analysis

Five controls each change one component of the final model and keep the data, recipe, and training budget, with three training seeds each (Fig. 5). They test the argument of

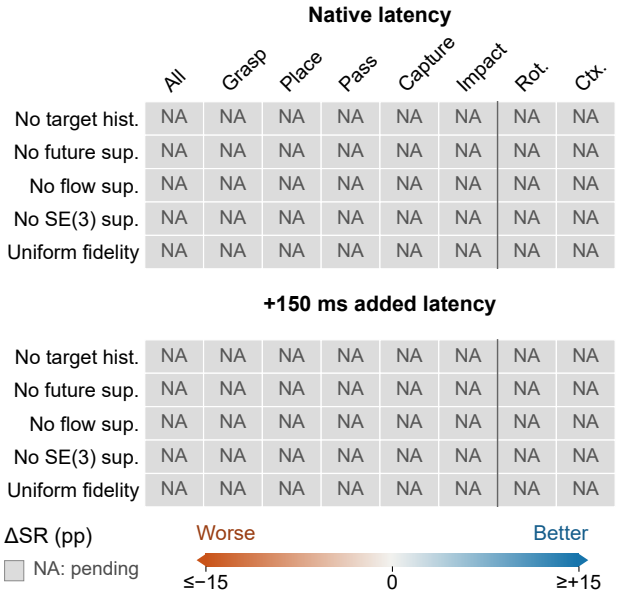


Figure 5. **Ablation effects by task (design preview; data pending).** ΔSR = SR(control) − SR(full), in percentage points; colors saturate at ±15. Rows: the five controls of Sec. 4.3; the full model reaches [XX.X]% SR with native latency and [XX.X]% with +150 ms. Rot./Ctx.: predeclared, overlapping rotation-sensitive/context-active groups (Table S3). Planned estimates average three seeds per scene and weight tasks equally, retaining `t3_phase`. Bold will mark unadjusted paired 95% scene-bootstrap intervals excluding zero, conditional on trained weights (Appendix B.4). Grey NA: no eligible paired results or estimates.

Sec. 3.3: the target history supplies the velocity on which e_v in Eq. S1 depends, future supervision shapes the representation without adding inference time, and the two geometric components and a uniform-fidelity forecast test the allocation of precision by role. We evaluate them with native latency and with 150 ms of added latency (per-seed results and a sensitivity analysis in Appendix C.4); since the forecast heads are skipped at deployment, all controls share the inference cost of SE(3)-WAM.

Target history. The no-history control removes the past target poses with their velocity and acceleration features and keeps the current image, the current target pose, and the robot history; unlike the other controls, it changes the input. Without the history, SR drops by [X.X] points with native latency and by [X.X] points with 150 ms of added latency, as Eq. S1 predicts: a longer lead amplifies the velocity error that the history removes. Because the history relies on pose estimates, we also perturb them; under synthetic correlated pose noise, SE(3)-WAM keeps [XX.X]% SR (Appendix C.5).

Future supervision. The no-future control removes all forecast losses, the SE(3), scene-flow, and future appearance targets, and keeps the current-target RGB reconstruction. Its inference is identical to that of SE(3)-WAM, yet SR drops by [X.X] points with native latency and by [X.X] points with 150 ms of added latency, most on [tasks with non-uniform target motion]. Offline, the learned SE(3) forecast is also more accurate than constant-velocity and constant-acceleration extrapolation of the same history (Table S9).

Allocation of precision. Three controls keep all auxiliary targets and change the geometric forecast. Removing the scene-flow supervision lowers SR by [X.X] points; because the flow also covers the target and the robot, and the appearance targets remain, this measures the contribution of coarse scene-motion supervision rather than of all context information. Removing the dedicated SE(3) supervision leaves the target with only the coarse flow that every part of the scene receives and lowers SR by [X.X] points. If precision follows the roles, the SE(3) supervision should matter most on tasks with a tight pose tolerance or a rotation-sensitive contact, and the flow supervision on tasks whose context constrains or drives the target. [Whether Fig. 5 shows this pattern; if it does not, state that the controls establish the contribution of each component but not the allocation by role.] The uniform-fidelity control replaces the SE(3) and scene-flow targets by [future latents of a frozen video VAE] of the whole frame at the same horizons, so that every part of the scene is forecast at one fidelity; at the same inference cost, it trails SE(3)-WAM by [X.X] points. [One-sentence summary of which components matter, and where.]

4.4. Real-Robot Experiments

Setup. We evaluate six tasks on a Franka arm ([model]) with [gripper model] and a fixed RealSense D435 RGB-D camera (Fig. 6; Appendix D): conveyor grasping (T1), placement into a moving receptacle (T2), turntable grasping (T3), moving toy-car grasping (T4), tennis-ball grasping (T5), and block grasping from a ramp cart (T6). Baselines are $\pi_{0.5}$ (VLA), DynamicWAM (dynamic WAM), and Diffusion Policy [12] (reactive). All methods train or fine-tune on the same 100 teleoperated demonstrations per task and run 20 trials per task with shared trial keys (Appendix D.3). Baselines receive RGB [and a wrist camera if their recipe uses one]; SE(3)-WAM also receives FoundationPose [44] target estimates, updated at every control-period boundary before its query (Appendix D.2). End-to-end latencies include tracking (Table 3).

Results. SE(3)-WAM succeeds in [XX.X]% of all trials, against [XX.X]% for [the strongest baseline], a paired difference of [X.X] points (95% interval [[X.X, X.X]]; Table 3). Its p95 latency, including tracking, is [XX] ms, [within one

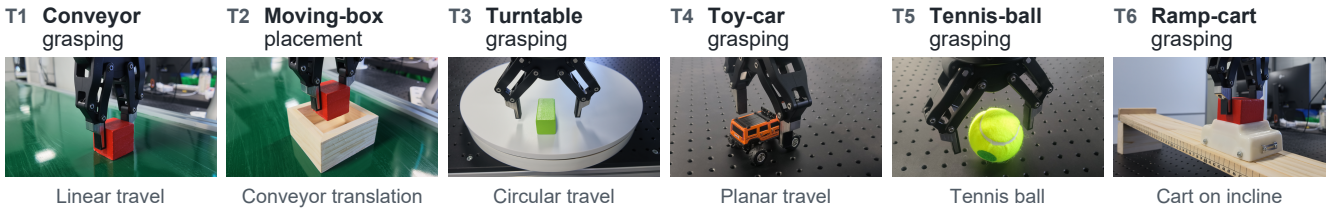


Figure 6. **Real-robot task setups.** Photographs of conveyor grasping (T1), placement into a moving box (T2), turntable grasping (T3), moving toy-car grasping (T4), tennis-ball grasping (T5), and grasping a block carried by a ramp cart (T6).

control period], whereas [name] needs [XX] ms. The margin is largest on [tasks], and the dominant failure of SE(3)-WAM is [failure mode] (Appendix D.4).

5. Conclusion

SE(3)-WAM spends precision where contact depends on it: a metric SE(3) history and forecast of the target and a sparse 3D flow of the whole scene, with the forecasts used only as training supervision. It reaches [XX.X]% success on Dynamic-14, [XX.X]% on DOMINO, and [XX]% on six real-robot tasks at a p95 latency of [XX] ms, improving on reactive, dynamic, and world–action policies. [The ablations show that the target history matters most under added latency and that future supervision improves control without changing the inference computation.] [State what the dedicated SE(3), scene-flow, and uniform-fidelity controls support: a better forecast design, or, if the task groups also agree, precision allocated by role.] Accounting for inference latency [state how success rates and relative performance differ from paused evaluation].

Limitations. SE(3)-WAM requires a specified rigid target and reliable estimates of its pose, and its forecast does not model how the robot's own contact changes the target's motion. Its target-trajectory head predicts a single future per horizon and does not represent uncertainty or multimodal motion, which an uncertainty-aware forecast could address.

LLM Usage Statement

We used OpenAI Codex to assist with literature review, code implementation and debugging, experiment execution, and language editing of the manuscript. Codex also assisted in developing plotting code and refining figure and manuscript layouts. The authors reviewed AI-assisted content, checked references against original sources, and validated the implementation through testing and experiments. The authors take full responsibility for the final manuscript, experimental results, and conclusions.

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A. Method and Training Details

A.1. Error Decomposition

This subsection bounds the error of a policy's estimate of the target pose at the time τ^{a} at which a command takes effect (Eq. (S5)), derives the rotation and context terms behind the task groups of Appendix B.1, and shows how the error scales with the lead and the motion. Let a policy act, explicitly or implicitly, on an estimate of the target pose at τ^{a} with position error e and rotation error θ, so that a contact point at lever arm r is off by at most $e + 2r\sin(\theta/2)$. Writing the target's motion over the lead $\Delta = \tau^{\mathrm{a}} - \tau_t$ as a constant velocity $\mathbf{v}$ plus a residual (acceleration, rolling, impacts), and the lead that the policy assumes as $\hat{\Delta}$, gives

$$e \le e_{\mathrm{loc}} + \hat{\Delta}\, e_v + \|\mathbf{v}\|\, e_\tau + e_{\mathrm{dyn}}, \tag{S1}$$

with the errors e_{loc} of the current position, e_v of the velocity, $e_\tau = |\hat{\Delta} - \Delta|$ of the timing, and e_{dyn} of the residual motion.

Translation. Write the true target position at the execution time as $\mathbf{p}(\tau^{\mathrm{a}}) = \mathbf{p}(\tau_t) + \mathbf{v}\Delta + \boldsymbol{\rho}$, with the velocity $\mathbf{v}$ at τ_t and the remaining motion $\boldsymbol{\rho}$, and the policy's estimate as $\hat{\mathbf{p}}(\tau^{\mathrm{a}}) = \hat{\mathbf{p}}(\tau_t) + \hat{\mathbf{v}}\hat{\Delta} + \hat{\boldsymbol{\rho}}$. Subtracting and adding $\mathbf{v}\hat{\Delta}$ gives

$$\begin{aligned}\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}}) = \big[\hat{\mathbf{p}}(\tau_t) - \mathbf{p}(\tau_t)\big] + (\hat{\mathbf{v}} - \mathbf{v})\hat{\Delta} \\ + \mathbf{v}(\hat{\Delta} - \Delta) + (\hat{\boldsymbol{\rho}} - \boldsymbol{\rho}),\end{aligned} \tag{S2}$$

and the triangle inequality yields Eq. (S1) with $e_v = \|\hat{\mathbf{v}} - \mathbf{v}\|$ and $e_{\mathrm{dyn}} = \|\hat{\boldsymbol{\rho}} - \boldsymbol{\rho}\|$. The bound holds for any split of the estimate into $\hat{\mathbf{v}}\hat{\Delta}$ and $\hat{\boldsymbol{\rho}}$, and e_{dyn} absorbs the higher-order part of the timing error. The residual $\boldsymbol{\rho}$ is the motion that constant-velocity extrapolation misses. Under a constant acceleration $\mathbf{a}$, $\boldsymbol{\rho} = \frac{1}{2}\mathbf{a}\Delta^2$, which an acceleration estimate from the history reduces; impacts, rolling contact, and friction changes leave a remainder that only a model of the dynamics can reduce.

Rotation and the contact point. Write the true orientation as $R(\tau^{\mathrm{a}}) = \mathrm{Exp}(\boldsymbol{\omega}\Delta + \boldsymbol{\eta})R(\tau_t)$, with the angular velocity $\boldsymbol{\omega}$ and the remaining rotation $\boldsymbol{\eta}$, and its estimate likewise. The rotation angle is bi-invariant, and Exp is 1-Lipschitz from $\mathbb{R}^3$ to SO(3), so for general rotations

$$\begin{aligned}\theta &= \big\|\,\mathrm{Log}(\hat{R}(\tau^{\mathrm{a}})R(\tau^{\mathrm{a}})^\top)^\vee\big\| \\ &\le \theta_{\mathrm{loc}} + \|\hat{\boldsymbol{\omega}}\hat{\Delta} + \hat{\boldsymbol{\eta}} - \boldsymbol{\omega}\Delta - \boldsymbol{\eta}\| \\ &\le \theta_{\mathrm{loc}} + \hat{\Delta}\, e_\omega + \|\boldsymbol{\omega}\|\, e_\tau + e_\eta,\end{aligned} \tag{S3}$$

with the current orientation error θ_{loc}, $e_\omega = \|\hat{\boldsymbol{\omega}} - \boldsymbol{\omega}\|$, and $e_\eta = \|\hat{\boldsymbol{\eta}} - \boldsymbol{\eta}\|$. A contact point at $\mathbf{c}$ in the object frame, with $\|\mathbf{c}\| = r$, is off by $(\hat{\mathbf{p}} - \mathbf{p}) + (\hat{R}R^\top - I)R\mathbf{c}$ at τ^{a}, and the operator norm of $\hat{R}R^\top - I$ is $2\sin(\theta/2)$, so its error is

$$e_{\mathrm{c}} \le e + 2r\sin(\theta/2) \le e + r\theta, \tag{S4}$$

with $e = \|\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}})\|$. A grasp or placement constrains several points spread over the body, such as the two finger contacts or the arms of a T-shaped part, so no single tracked point removes the term $r\theta$: the orientation error enters the contact scaled by the lever arm, and the target needs its full SE(3) state.

Context. Let m be the largest margin that a feasible path can keep between the robot and the carried object and the context it passes, measured to the predicted positions of the context at τ^{a}. If the executed path keeps this margin, a collision requires a context error of at least m, so a coarse estimate of the surroundings suffices when $m \gg \varepsilon$; the no-scene-flow control of Sec. 4.3 tests this coarse supervision directly. A driver, such as a conveyor, a ramp, or a striker, acts on the contact through the motion it imparts to the target, which enters Eq. (S1) through the target's own terms. The SE(3) history and forecast represent this motion precisely, so the driver needs no precise state of its own.

Table S1. Architecture of SE(3)-WAM. Heads are skipped at deployment.

Module	Configuration
Visual encoder	frozen MAE ViT-B/16, 224×224 input, 196 patch tokens
Target-state tokenizer	$28 \to 1024 \to 1024$ GELU MLP, additive relative-time embedding
Robot-state tokenizer	linear, $7 \to 1024$
Tokens	212 scene (196 patch, 8 target-state, 8 robot), 16 summary, 8 action
Backbone	24 layers, width 1024, 16 heads, shared attention, SiLU FFN 4096 per stream, 504.0M parameters
Pose head	linear, mean summary state $\to 8 \times 6$
Flow head	linear per patch token, $8 \times (3{+}1)$
Appearance heads	linear per patch token: depth and moving-surface RGB as 16×16 pixels, 1024-d DINOv3 ViT-L/16 features
Target reconstruction	$1024 \to 256 \to 3072$ GELU MLP with sigmoid, 32×32 RGB
Action stream	flow matching, adaptive layer norm on the flow time, 5 Euler steps

A.2. Architecture

Table S1 lists the configuration of every module, in the order of the data flow described below.

Scene tokens. The frozen MAE ViT-B/16 [17] encodes the agent-view RGB into 196 patch tokens; we selected it on development scenes (Appendix E.4). Each patch token adds a learned view embedding, shared by all patches in our single-view setting, and linear embeddings of its mean

valid metric depth and of the base-frame position of its back-projected center pixel. These positions also drive 3D rotary embeddings of the patch queries and keys, which no other token receives, and the same center points are the source points of the sparse 3D flow. Past images never enter the model.

State tokens. The 8 target-state tokens sample the offsets $-350, -300, \ldots, 0$ ms. Each is a 25-dimensional descriptor $(\mathbf{p}, \log R, \Delta\mathbf{p}, \log \Delta R, \Delta t, \mathbf{v}, \boldsymbol{\omega}, \mathbf{a}, \boldsymbol{\alpha})$ with dimensions $(3, 3, 3, 3, 1, 3, 3, 3, 3)$: slots 0–6 describe the seven intervals between consecutive offsets, with velocities from interval displacements over the elapsed time and accelerations from adjacent velocities over the spacing of the interval centers, and slot 7 holds the current absolute pose in the base frame. We standardize each dimension with training statistics, zero invalid dimensions, and append three validity flags (pose, interval, and acceleration); a slot without any valid feature is masked out of attention. On the real robot, the tracker updates the pose at every control-period boundary (Appendix D.2), so the history keeps the same 50 ms spacing. The robot tokens hold the end-effector position, rotation vector, and measured gripper opening at the same offsets, and 16 learned embeddings form the summary tokens. The policy has no language token.

Backbone and mask. Each layer applies one attention over all tokens with query, key, value, and output projections shared by both streams, followed by separate SiLU feed-forward networks for the world and the action tokens. The mask of Sec. 3.4 gives a blocked edge the logit $-\infty$ in every layer and head, in training and at deployment, so the world states are computed once per query and reused across the Euler steps.

Forecast and auxiliary heads. All heads are readouts of the final world states, trained jointly with the action stream and skipped at deployment; no token reads their outputs. The pose head reads the mean summary state and is trained with $\mathcal{L}_P$ (Eq. 3). The flow head gives each patch point's displacement and visibility logit at every horizon, trained with $\mathcal{L}_F$ (Eq. 2) and the scale $s = 0.7154605$ m; the visibility loss has weight 1 in simulation, and points that are valid but invisible still supervise their displacement. The appearance heads predict at every horizon the metric depth with a squared error on valid pixels, the RGB of surfaces that move by more than 1 mm, including the robot, with a squared error, and the DINOv3 features [35] with a squared error after channel-wise layer normalization. The target-reconstruction head recovers the current target's RGB region from the mean summary state with an absolute error. These four losses form $\mathcal{L}_{\text{aux}}$.

Action stream. Each of the 8 action tokens embeds a seven-dimensional command, a base-frame translation increment, a left-relative rotation vector, and a gripper command; the increments refer to the measured end-effector pose at application. At deployment, we integrate 5 Euler steps, reverse the normalization, decode the rotation vectors, and clip each axis to 0.05 m and 0.5 rad; the gripper follows the sign of its command. Appendix B.3 specifies how the chunks are executed.

A.3. Training on Dynamic-14

Each control step of a demonstration anchors one window of 8 query positions with their own causal histories and no attention across positions, which gives 1,324,374 windows from the 11,200 demonstrations; 350 further episodes form the validation split. For the flow labels, each patch-center pixel is back-projected with the rendered depth and attached to the local frame of the MuJoCo geometry rendered at it; the ground-truth rigid transforms of that geometry carry the same surface point to the future horizons, which gives its cumulative displacement in the base frame. A future point is visible if it lies in front of the camera and inside the image and its projected pixel renders the same geometry at a depth within $\max(1\,\text{mm}, 0.001z)$ of the point's depth z. Command j of the chunk for query t is labeled with the demonstration command issued at $\tau_t + j\Delta_a$, since d_{task} applies only at execution (Eq. (S5)), and commands are normalized with the mean and standard deviation of the valid commands of the training split. Each query position draws 8 noise–time pairs for its chunk and averages their losses, which reduces the variance of the action gradient without enlarging the batch, and each loss is normalized by its count of valid labels in the global batch. The DINOv3 features of future frames are cached once per episode. Table S2 lists the recipe.

Table S2. Training hyperparameters on Dynamic-14.

Item	Value
Optimizer	AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 10^{-4}
Learning rate	2×10^{-4}, 5,174 warm-up updates from 0, cosine decay to 0
Batch, updates	global batch 512 windows, 20 epochs (51,740 updates)
Loss weights	$\lambda_P = \lambda_F = \lambda_R = 0.1$, $\lambda_{\text{aux}} = 0.1$ per target
Action noise	8 noise–time pairs per query position
Precision	trainable weights FP32, frozen encoder BF16, gradient clipping at 0.1
Inference	FP32, world states cached within a query
Hardware	[N] RTX PRO 6000 GPUs, [X] hours per run

A.4. Pretraining on DROID and Fine-Tuning

Data. The real-robot base trains on DROID [20], recorded at 15 Hz, with the 3D point tracks and camera calibrations that PointWorld released for it [19], matched to the raw episodes by exact agreement of robot states, commands, and camera identity. It uses the first exterior camera with its depth. Of 42,935 matched episodes, 42,087 yield 598,889 anchor windows, 538,740 from 37,874 training and 60,149 from 4,213 validation episodes. Each anchor expands into 8 consecutive query positions; the anchor carries the flow and DINOv3 labels, and the others carry action labels only. All times are capture timestamps, and the flow and DINOv3 targets keep PointWorld's record times. We keep the 96.88% of action slots with an issued command, without a filter on speed, so that the base is not biased toward slow motion.

Objectives. DROID provides no object poses, so the base has no target-state tokens or pose head. It reads the current RGB-D frame and the robot history (204 scene tokens) and trains the action loss and the sparse 3D flow with weight 1 and the future DINOv3 features with weight 0.1; PointWorld records only visible positions, so the visibility loss is off. Most DROID objects move only when the robot moves them, so the base mainly learns metric geometry and the motion caused by the robot's own actions; the exogenous motion of our tasks and the target pose enter during fine-tuning.

Flow labels. Each patch of the 14×14 grid follows one PointWorld track: among the tracks that start inside the patch, are initially visible, and have a valid depth, the one with the largest image motion consistent with its 3D displacement, and otherwise the one nearest to the patch center, so that moving objects missed by a fixed center point are covered. The label is the track's displacement from its start; 47.05% of the point–horizon slots carry a label.

Recipe and fine-tuning. The base has 48 layers at width 1024 (1.02B trainable parameters) and trains for 20 epochs (21,060 updates, 2,106 warm-up updates) at a global batch of 512 on 64 GPUs, with the optimizer and precision of Table S2. Each real-robot policy starts from the base and is fine-tuned on the demonstrations of its task [updates, learning rate, and batch size]; the target-state tokens and the heads that the base lacks are freshly initialized. No real-robot policy uses simulation weights, and the base sees no simulation data.

B. Dynamic-14 and Evaluation Protocol

B.1. Benchmark Design

Dynamic-14 comprises fourteen tasks in five families (Fig. 3; Sec. 4.1), which together cover translation, rotation, temporary occlusion, constrained passage, and abrupt changes of the target's motion. We will release the tasks, the accepted scenarios, the success checks, and the demonstrations.

Environment. The tasks run in MuJoCo 3.11.0 and robosuite 1.5.2 [42, 62] with a Franka Panda arm and a parallel gripper, driven by robosuite's operational-space pose controller at a control period of 50 ms and a physics step of 2 ms. A fixed third-person agent-view camera at $(0.68, -0.58, 1.62)$ m with a 40° vertical field of view renders 224×224 RGB-D images. An action is an end-effector increment in the robot base frame, a translation and a rotation vector, together with a gripper command; the shared action adapter clips each command to 0.10 m per translation axis and 0.5 rad per rotation-vector component.

Success and termination. Table S3 summarizes the success condition of each task. We check them at every physics step, an interrupted continuous condition resets its counter, and an episode ends at its first terminal event: success, a failure such as unsafe contact, leaving the workspace, or a drop, or the time limit.

Roles and task groups. Table S3 names the target of each task. A moving receptacle is the target, since its pose decides containment; in the passage tasks, the wire entry, which the wand must match within 30 mm, is the target, and the rest of the wire is context that no contact may touch. The two task groups of Fig. 5 were fixed before the ablations from the terms of Appendix A.1: a task is rotation-sensitive ($r\theta$) if its target rotates during the lead and its orientation matters at contact, which excludes the balls, and context-active if it has an obstacle clearance m or a driver (a conveyor, a ramp, or a striker).

B.2. Scenarios, Demonstrations, and Splits

Scenarios. Each scene samples the target's initial state and motion parameters uniformly within task-specific ranges, and discrete choices with equal probability: besides the speeds of Table S3, harmonic amplitudes of up to 35 mm at 0.25–0.7 Hz, yaw rates of up to 1.2 rad/s, wire rotation rates of 2.5–7.5°/s, and ramp slopes of 5–15° with friction set per object. An occluded scene is accepted only if the target is visible for at least eight initial frames, fully occluded for 0.5–3.0 s, and visible again for at least three frames. We keep the scenes that pass physical-feasibility checks and replay the stored accepted scenes at evaluation.

Demonstrations. The expert is a task-specific closed-loop state machine with privileged simulator state: the target's current pose and linear and angular velocity, the robot state, the task geometry, and true contact, grasp, and stage events.

Table S3. **The fourteen Dynamic-14 tasks.** Target: the body whose pose the policy receives (Sec. 3.1). Speed: target speed range (capture tasks: at the ramp exit). Every task requires its stages in order without a terminal failure; a grasp requires bilateral fingerpad contact, a lift of at least 0.05 m, and a 0.5 s hold. Limit: episode time limit. Groups of Fig. 5: R, rotation-sensitive; C, context-active (Appendix B.1). The released task configurations contain the exact thresholds and scenario ranges.

Task (code)	Target	Speed (m/s)	Success	Limit (s)	Group
Harmonic grasp (grasp)	block	0.03–0.20	Grasp the block moving harmonically and yawing, lift, and hold.	8	R
Conveyor grasp (t1_belt)	part	0.04–0.10	Grasp the part on the conveyor, lift, and hold.	8	C
Translating grasp (t1_xyz)	part	0.04–0.15	Grasp the translating part, lift, and hold.	8	–
Rotating grasp (t1_so3)	part	0.04–0.15	Grasp the translating and rotating part ($\|\omega\| \leq 1.2$ rad/s), lift, and hold.	8	R
Occluded grasp (t1_occ)	part	0.04–0.10	Grasp the part on the conveyor, which a canopy hides for 0.5–3.0 s, without unsafe contact with the canopy; lift and hold.	8	C
Moving receptacle (t2_trans)	receptacle	0.03–0.06	Place the held object fully inside the moving receptacle and release it to rest.	10	–
Rotating receptacle (t2_se3)	T-shaped receptacle	0.03–0.06	Place the held T-shaped part fully inside the translating and rotating T-shaped receptacle and release it to rest.	10	R
Wire passage (t3_phase)	wire entry	[X]	Align the held wand's loop with the rotating wire within 30 mm, then pass it along the wire to the end without contact.	21	R, C
Full wire passage (t3_full)	wire entry	[X]	Grasp the wand, complete the passage, exit the wire, and release the wand.	32	R, C
Ball capture (t4_sphere)	ball	[0.4–0.7]	Catch the ball after it leaves the ramp and before it touches the tray, lift, and hold.	6	C
Ball capture on table (t4_sphere_tabletop)	ball	[0.4–0.7]	Grasp the ball while it still rolls on the table after the ramp, lift, and hold.	6	C
Slider capture (t4_slider)	slider	[0.4–0.8]	Grasp the block while it still slides on the table after the ramp, lift, and hold.	6	C
Can capture (t4_can)	can	[0.4–0.7]	Grasp the can while it still rolls on the table after the ramp, before it tips over, lift, and hold.	6	R, C
Post-impact grasp (t5_replan)	part	0.12–0.14	After a striker deflects the part on the conveyor, grasp it along its grasp axis, lift, and hold.	6	R, C

At every control period it, for example, extrapolates the target over a short horizon and tracks, descends, closes, and lifts in the grasping tasks, predicts the interception point of a rolling or flying object under gravity in the capture tasks, and follows the known wire path with phase compensation from the current angle and angular velocity in the passage tasks. Its commands are clipped to 0.05 m and 0.5 rad per axis. A demonstration is accepted if its scene is feasible, the expert succeeds, an independent replay of the recorded actions reproduces the episode, and its path, action changes, and jerk pass numerical quality checks. The training pool of 11,200 demonstrations, 800 per task, is drawn from the accepted ones. We record them at 20 Hz with camera images, depth, robot states, issued commands, and simulator ground truth for target poses and scene geometry; the learned policies receive only the inputs of Sec. 3.

Splits. We select configurations on 1,400 development scenes (100 per task). All Dynamic-14 results use a separate held-out set of 1,400 scenes, generated with the same ranges and new source seeds that appear in no training, validation, or development episode; the first 100 feasible scenes in increasing seed order form each task's set, which all methods share.

B.3. Real-Time Evaluation

Executor. All methods share one executor on the 50 ms control clock. Command j of query t is indexed by its intended issue time $\tau_t + j\Delta_a$; if executed, it takes effect at

$$\tau^{\mathrm{a}}_{t,j} = \tau_t + j\Delta_a + d_{\mathrm{task}}, \tag{S5}$$

where $d_{\mathrm{task}} = 50$ ms for the post-impact grasp, which applies each command 50 ms after it is issued, and $d_{\mathrm{task}} = 0$ for all other tasks. The total latency L_t of a query is its inference latency under the protocol plus any added latency. The executor counts latency in whole control periods, as RTC does [7]: a query hands over its chunk at the first control-period boundary at or after it ends, $k_t = \lceil L_t/\Delta_a \rceil$ periods after capture, and execution continues from command k_t, the one meant for that boundary. There the policy observes again and starts the next query; until then, the executor continues the previous chunk, so a policy replans as often as its latency allows. A method's chunk holds its native number H_m of commands (Table S4), and a chunk with $k_t \geq H_m$

has expired entirely and is dropped. With $L_t = 0$, the next query starts one control period after capture. Before the first chunk of an episode arrives, and whenever a chunk runs out, the controller holds the current arm target and gripper command while physics advances; no command is repeated. Every executed command of SE(3)-WAM takes effect at most $(H-1)\Delta_a + d_{\text{task}}$ after capture, within its forecast horizons (Sec. 3.3).

Protocols. Under the real-time protocol, physics advances during each query by the method's native latency, drawn per query from its latency distribution measured on one RTX 5090. With paused physics, the simulation waits for inference; with equal latency, every method is charged the p95 latency of SE(3)-WAM; with original executors, each baseline keeps the execution scheme of its paper, such as temporal aggregation for ACT or streaming for DynamicVLA, at its native latency. The added-latency sweep of Table S7 adds 0, 50, 100, 150, or 200 ms after inference under the real-time protocol, and physics advances during it. The native-latency rollouts of ACT and Diffusion Policy ran on RTX 5090 and all other rollouts on RTX PRO 6000; every rollout replays the latency measured on RTX 5090, so the rollout hardware does not affect the results.

Latency measurement. We time all methods on one RTX 5090 (32 GB), the GPU of our robot, with exclusive use and batch size 1. Each method runs three rounds of thirty warm-up and 300 measured calls over three observation sequences of eight frames, with CUDA synchronization at call boundaries and a rotated method order, in the precision of its release; SE(3)-WAM uses FP32. A timed query starts from the newest observation, with the history already maintained, and ends when its first command is available. It includes all computation that needs the newest frame: preprocessing, image encoding, the online flow updates and motion descriptors of PUMA and DynamicWAM, sampling, and action decoding, and on the robot also tracking. For SE(3)-WAM, we time the forecast heads separately, although deployment skips them. This time-to-action is reported without rounding; the handover time $\ell_t = k_t\Delta_a$ rounds it up to the control period, and the action age of a command is its execution time minus the capture time. A query misses its deadline when its time-to-action exceeds 50 ms. Appendix C.2 reports the measured values.

B.4. Metrics and Statistics

The task-balanced success rate is

$$\mathrm{SR}(m) = \frac{100}{14}\sum_{k=1}^{14}\frac{1}{S_m n_k}\sum_{s=1}^{S_m}\sum_{i=1}^{n_k} y_{mski}, \qquad \text{(S6)}$$

for binary outcomes y_{mski} of method m, trained with seed s, on scene i of task k, with $S_m = 3$ training seeds for our variants and $S_m = 1$ for the baselines. The *position error at first contact* e_p is the distance of the end effector from the expert's contact frame on the target. In the two placement tasks, which start with the object held, the contact is that of the held object with the receptacle, and the contact frame is the end-effector pose of the expert's placement. e_p averages over the 13 tasks with a contact, excluding `t3_phase`, and over the episodes that make contact; failures without contact stay in the SR denominator.

Table S4. **Baseline training.** All methods train on the same 11,200 demonstrations with the official recipe of each method and run under the shared executor of Appendix B.3. Updates × batch: training budget. H: commands per returned chunk. Params: trainable parameters, frozen encoders excluded. DOM ckpt.: the released DynamicVLA checkpoint trained on DOM. OpenWAM-α generates video and actions jointly with video decoding disabled.

Method	Initialization	Updates×batch	H	Params
ACT [59]	scratch	50,000 × 32	100	[X]
DP [12]	scratch	50,000 × 32	8	[X]
$\pi_{0.5}$ [5]	`pi05_base`	30,000 × 256	10	[X]
$\pi_{0.5}$+SE(3)	`pi05_base`	30,000 × 256	10	[X]
DynamicVLA [47]	DOM ckpt.	50,000 × 44	20	[X]
PUMA [14]	official	[X]×[X]	[X]	[X]
OpenWAM-α [43]	official	34,490 × 192	32	[X]
DynamicWAM [28]	official	40,000 × 128	16	[X]
DynamicWAM+SE(3)	official	40,000 × 128	16	[X]
SE(3)-WAM	scratch	51,740 × 512	8	523.8M

Training seeds and intervals. The full model and the five ablation controls use training seeds 0, 1, and 2, and each baseline uses seed 0. We report each training seed and the cross-seed mean and standard deviation (Appendix C.4), and for the ablations also the comparison without the phase-only wire passage (`t3_phase`). For paired 95% intervals, we average the seeds per scene and bootstrap held-out scenes within each task, jointly across methods. These intervals quantify scene uncertainty for the trained weights, and we treat differences whose interval lies within ±2 points as ties.

B.5. Baselines and Ablation Controls

Common conditions. Each baseline keeps its official architecture, chunk length, initialization, and training recipe (Table S4), trains once with seed 0 on the same 11,200 demonstrations, whose commands we map to its action space through the same Cartesian controller, and is configured on development scenes. All methods share the agent-view camera and receive the further inputs of Table 1; methods whose recipes expect wrist views receive masked or black wrist inputs.

SE(3)-augmented baselines. $\pi_{0.5}$+SE(3) and DynamicWAM+SE(3) receive the eight target-state tokens of Ap-

pendix A.2 from the same pose source, with the same validity flags and masking. A two-layer MLP maps the tokens to the model's width. For $\pi_{0.5}$, we append them to the prefix after the image tokens, where the action expert attends to them; for DynamicWAM, they join its kinematic descriptors. The output layer of the MLP is zero-initialized, and training follows the base model's recipe.

Ablation controls. The five controls of Sec. 4.3 change one component each of the configuration of Table S2 and keep the data, the recipe, the update budget, the executor, and the training seeds 0, 1, and 2. *No target history* removes slots 0–6 of the target-state tokens, which carry the past poses with their velocity and acceleration features, and keeps slot 7 with the current pose (Appendix A.2). *No future supervision* removes $\mathcal{L}_P$, $\mathcal{L}_F$, and the three future appearance losses with their heads, and keeps the reconstruction of the current target region. *No scene-flow supervision* removes $\mathcal{L}_F$ and the flow readout, and *no dedicated SE(3)* removes $\mathcal{L}_P$ and the pose head; both keep the target-state tokens and all appearance targets. The *uniform-fidelity forecast* replaces $\mathcal{L}_P$ and $\mathcal{L}_F$ by [a forecast of future latents of a frozen video VAE of the whole frame at the same horizons, its readout, and its loss weight]. Since every forecast head is skipped at deployment, only the no-history control changes the inputs at inference.

C. Additional Simulation Results

C.1. Per-Task Results

Table S5 breaks the held-out Dynamic-14 results of Table 1 down by task for every method, with the paired interval of SE(3)-WAM against the strongest other method on each task; its last two rows give the position error at first contact e_p and the number of episodes with a contact (Appendix B.4). With its native 100-step DDPM sampler, every chunk of Diffusion Policy has expired when it arrives, so none of its commands is executed. [Tasks on which every method fails or a baseline outperforms SE(3)-WAM, and the dominant failure of SE(3)-WAM.]

C.2. Evaluation Protocols and Latency

Table S6 evaluates every method under three further protocols. Paused physics shows how the relative performance changes when the cost of computation is ignored, equal latency compares all methods at the p95 latency of SE(3)-WAM, and the original executors check whether the shared executor removes a capability that a baseline relies on. With paused physics, the difference between SE(3)-WAM and [name] is [$\pm$X.X] points, and under equal latency it is [$\pm$X.X] points. The original executors change the SR of the baselines by at most [X.X] points. Table S7 lists the available success rates and pending cells at each added latency. Fig. 4 reads this table directly and shows all methods with accepted measurements, including zero success rates.

Timing. Table S8 breaks the time-to-action of SE(3)-WAM on one RTX 5090 down by stage, measured as in Appendix B.3; Table 1 gives the p95 latency of every method and Table S4 the parameter counts. On the robot, we report the p50, p95, and maximum action age and the fraction of all queries of the test trials that miss the 50 ms deadline, together with the peak memory.

C.3. Results on DOMINO

On DOMINO [14], we compare SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of its clean level-1 setting. All methods use the benchmark's demonstrations, scenes, and native evaluator, one training seed, and 50 episodes per task and method. We report SR and the manipulation score (MS), which scales route completion by penalties for leaving the workspace or view and for collisions (Table 2). SE(3)-WAM uses domain-specific normalization and reads the simulator pose of the target named by the instruction; slots without a valid pose are masked as in Appendix A.2. The RGB-based baselines do not receive this pose. [Summarize the DOMINO results.]

C.4. Final Ablations and Seed Stability

Table S10 reports the five controls of Sec. 4.3 (Appendix B.5) for each training seed, with the cross-seed mean and standard deviation and the paired difference from the full model, with native latency and with 150 ms of added latency. The last two columns repeat the paired differences without the phase-only wire passage (`t3_phase`). On development scenes, three training seeds of the same configuration differed by up to [7.4] points in SR, mostly on this task, whose success ranged from [0] to [71]% across seeds (Appendix B.4). The two task groups of Fig. 5 are listed in Appendix B.1. [Whether any conclusion of Sec. 4.3 changes for an individual seed or without `t3_phase`.]

C.5. Robustness to Pose Noise

With frozen weights, we evaluate three synthetic perturbations of the target pose: nominal temporally correlated noise, twice its amplitude, and nominal noise with contiguous missing observations. The nominal process combines a fixed per-episode bias and a stationary correlated component, each with per-axis standard deviations of 5 mm and 1°, with a correlation time of 200 ms. Dropouts last 200 ms, separated by geometrically distributed valid gaps with a mean of 3.8 s, giving a long-run missing fraction of 5%. The perturbation applies to the current and past estimates before the history features are computed.

Fig. S1 evaluates the final model under these perturbations, on the same held-out scenes at zero added latency and with all three training seeds. Relative to clean poses, SR

Table S5. **Per-task held-out success** (%) on Dynamic-14 for every method of Table 1, mean of three training seeds for our variants and one for baselines, real-time protocol. DP: Diffusion Policy. DVLA: DynamicVLA. OWAM: OpenWAM-α. DWAM: DynamicWAM. +: with our SE(3) history tokens. σ: SE(3)-WAM with nominal synthetic correlated pose noise (Appendix C.5). Δ: SE(3)-WAM minus the strongest other method on that task, with its 95% paired interval. The last two rows give the position error at first contact e_p (mm), averaged over the 13 tasks with a contact, and the number n of episodes with a contact (Appendix B.4); –: no 13-task mean. Task codes are listed in Table S3. Shading of Δ: blue for higher SR, orange for lower, saturating at 15 points. Blue/bold and light blue rank the reported values only; pending entries are excluded.

Task	ACT	DP	$\pi_{0.5}$	$\pi_{0.5}$+	DVLA	PUMA	OWAM	DWAM	DWAM+	σ	**Ours**	Δ [95% CI]
Harmonic grasp	**48.0**	0.0	—	—	—	—	—	—	—	—	—	—
Conveyor grasp	**46.0**	0.0	—	—	—	—	—	—	—	—	—	—
Translating grasp	**18.0**	0.0	—	—	—	—	—	—	—	—	—	—
Rotating grasp	**21.0**	0.0	—	—	—	—	—	—	—	—	—	—
Occluded grasp	**32.0**	0.0	—	—	—	—	—	—	—	—	—	—
Moving receptacle	**26.0**	0.0	—	—	—	—	—	—	—	—	—	—
Rotating receptacle	**0.0**	**0.0**	—	—	—	—	—	—	—	—	—	—
Wire passage	**99.0**	0.0	—	—	—	—	—	—	—	—	—	—
Wire passage, full	**0.0**	**0.0**	—	—	—	—	—	—	—	—	—	—
Ball capture	**36.0**	0.0	—	—	—	—	—	—	—	—	—	—
Ball capture, table	**71.0**	0.0	—	—	—	—	—	—	—	—	—	—
Slider capture	**57.0**	0.0	—	—	—	—	—	—	—	—	—	—
Can capture	**90.0**	0.0	—	—	—	—	—	—	—	—	—	—
Post-impact grasp	**44.0**	0.0	—	—	—	—	—	—	—	—	—	—
Task-balanced	**42.0**	0.0	—	—	—	—	—	—	—	—	—	—
e_p (mm)	32.2	–	—	—	—	—	—	—	—	—	—	–
n	1109	6	—	—	—	—	—	—	—	—	—	–

Table S6. **Protocol controls** on held-out Dynamic-14 scenes (SR, %) under the protocols of Appendix B.3; real time is the protocol of Table 1. Late: queries in real time whose time-to-action exceeds the 50 ms control period (%). Age: p95 action age in real time (ms); –: no command executed. Blue/bold and light blue rank the reported values only; pending entries are excluded.

Method	Paused	Equal latency	Original executor	Real time	Late (%)	Age p95
ACT	**40.9**	—	—	**42.0**	**0.0**	50
Diffusion Policy	33.1	—	—	0.0	100.0	–
$\pi_{0.5}$	—	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	—	—	—	—	—	—
DynamicVLA	—	—	—	—	—	—
PUMA	—	—	—	—	—	—
OpenWAM-α	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—
DynamicWAM + SE(3)	—	—	—	—	—	—
SE(3)-WAM	—	—	–	—	—	—

Table S7. **Success versus added latency** on held-out Dynamic-14 scenes (SR in %, real-time protocol). The 0 and 150 ms columns equal the Native and +150 ms columns of Table 1. Blue/bold and light blue rank the reported values only; pending entries are excluded.

	Added latency (ms)				
Method	0	50	100	150	200
ACT	**42.0**	—	—	**24.1**	—
Diffusion Policy	0.0	—	—	0.0	—
$\pi_{0.5}$	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	—	—	—	—	—
DynamicVLA	—	—	—	—	—
PUMA	—	—	—	—	—
OpenWAM-α	—	—	—	—	—
DynamicWAM	—	—	—	—	—
DynamicWAM + SE(3)	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—

changes by [X.X], [X.X], and [X.X] points under nominal correlated noise, twice its amplitude, and missing observations.

C.6. Forecast Diagnostics

Table S9 compares the learned SE(3) forecast of the final model offline with constant-velocity and constant-acceleration extrapolation from the same pose history.

Table S8. **Latency breakdown** of one query of SE(3)-WAM on one RTX 5090 at batch size 1 (ms, p50 / p95), and action age on the robot.

Stage	p50	p95
Pose tracking (per frame, robot)	—	—
Image encoding	—	—
World stream (once per query)	—	—
Action stream (5 Euler steps)	—	—
Time-to-action (total)	—	—
Forecast heads (training only, for reference)	—	—
Action age at execution (robot)	—	—
Queries over 50 ms (robot, %)	—	

Table S9. **Offline prediction errors.** Target-position error at 400 ms (mm) on the same held-out observations. CV and CA are analytic extrapolations from causal pose history; the learned head reads frozen full-model states.

	Const. vel.	Harmonic	Roll/slide	Impact
CV extrapolation	—	—	—	—
CA extrapolation	—	—	—	—
Learned pose head	—	—	—	—

Table S10. **Final ablations by training seed** on held-out Dynamic-14 scenes (Appendix C.4). S0–S2: SR (%) of training seeds 0–2; mean ± s.d. over seeds; Δ: control minus SE(3)-WAM (points) with its paired 95% interval. Last two columns: Δ without the phase-only wire passage (`t3_phase`), 13 tasks. Shading of Δ: blue for higher SR, orange for lower, saturating at 15 points.

	Native					+150 ms					Δ w/o `t3_phase`	
Variant	S0	S1	S2	Mean ± s.d.	Δ [95%]	S0	S1	S2	Mean ± s.d.	Δ [95%]	Native	+150 ms
SE(3)-WAM *	—	—	—	—	–	—	—	—	—	–	–	–
No target history	—	—	—	—	—	—	—	—	—	—	—	—
No future supervision	—	—	—	—	—	—	—	—	—	—	—	—
No scene-flow supervision	—	—	—	—	—	—	—	—	—	—	—	—
No dedicated SE(3)	—	—	—	—	—	—	—	—	—	—	—	—
Uniform-fidelity forecast	—	—	—	—	—	—	—	—	—	—	—	—

Figure placeholder: Fig. S1 — pose noise

x: clean poses, nominal correlated noise, twice its amplitude, and nominal noise with missing observations; y: SR of SE(3)-WAM per task family, frozen weights, real-time protocol; paired intervals over matched scenes.

Figure S1. **Robustness to pose noise.** SR of SE(3)-WAM under synthetic perturbations of the target pose, relative to clean poses.

D. Real-Robot Experiments

D.1. Platform and Tasks

The robot is a Franka arm ([model]) with [gripper model] and a fixed RealSense D435 RGB-D camera, commanded through [a timestamped Cartesian interface] at [control frequency]. The camera records RGB-D at [resolution] and [frame rate], and we calibrate its extrinsics by [procedure] before each session. SE(3)-WAM uses this fixed third-person view; baselines whose recipes use a wrist view also receive [wrist camera]. All policies run on one RTX 5090. The task devices share this platform and are installed in turn, one task at a time (Fig. S2). Table S11 defines the tasks. The turntable target of T3 is a block with unequal sides that must be grasped across its short sides, so that its orientation at contact matters, and the toy car of T4 runs at a fixed throttle setting.

D.2. Pose Tracking and Labels

Online tracking. Tracking is marker-free. Each object instance has one metric mesh with a fixed origin and axes, prepared once by [reference capture] and shared by all demonstrations and trials; a ball is defined by its center and radius. In the first frame, Grounding DINO [25] localizes the object named by the task, SAM 2.1 [32] segments it, and FoundationPose [44] registers its pose on the mesh. Afterwards, Cutie [11] propagates the mask in every frame, and a 6D Kalman filter predicts the pose; its position is corrected by the center of the mask's bounding box and by the median depth inside the mask, and one refinement step of FoundationPose turns the corrected pose into the estimate, which is the filter's next measurement. We run the TensorRT implementation of FoundationPose in FP16 with CUDA Graphs. An empty mask makes the estimate invalid; the previous pose is not carried forward, and [re-initialization after a loss].

Timing. The tracker runs on the control clock. At every control-period boundary, it updates the pose on the newest frame captured at or before the boundary, so each estimate is at most one camera-frame period old, and it keeps the capture time of its frame; velocities and accelerations of the

Figure placeholder: Fig. S2 — real-robot setup

(a) Left, photo of the shared platform from behind the robot, with one task device installed: the third-person RGB-D camera, the robot, and the wrist camera if a baseline uses one, each labeled. **(b) Right, six small top-down schematics to scale, one per task (T1–T6):** the robot base frame with its axes, the camera position and viewing direction, the installed device with its footprint, the start or release region of the target and the range of its motion directions, the grasp region, and key dimensions in cm. **Rules:** real photo, unretouched; schematics from the calibrated extrinsics and measured device positions; no repetition of the task photos of Fig. 6.

Figure S2. **Real-robot setup.** (a) The shared platform. (b) Top-down layout of each task: the device, the start or release region and motion directions of the target, and the grasp region.

Table S11. **Real-robot tasks.** Target: the tracked object. Trial key: what each of the 20 test keys fixes. Speed: target speed measured from the recorded RGB-D of all test trials (m/s, mean ± s.d.). A grasp holds the object between both fingers. Each task has 100 training and 20 validation demonstrations.

Task	Target	Motion; trial key	Speed	Success	Limit (s)
T1	Block	Belt conveyor; [placement on the belt, belt speed]	—	Grasp the block and lift it by [X] cm.	[X]
T2	Open box	Belt conveyor; [box placement, belt speed]; the block starts in the gripper	—	Release the block into the box; it stays inside [until the box leaves the workspace].	[X]
T3	Block, unequal sides	Turntable; [radius, initial angle and yaw of the block, angular speed]	—	Grasp the block across its short sides and lift it by [X] cm.	[X]
T4	Toy car	Remotely driven at a fixed throttle; [start pose, heading]	—	Grasp the car and lift it by [X] cm.	[X]
T5	Tennis ball	[ball-motion mechanism, release position and motion setting]	—	Grasp the tennis ball and lift it by [X] cm.	[X]
T6	Block on cart	Cart translates down an inclined rail; [slope, release point]	—	Grasp the block and lift it by [X] cm.	[X]

target-state tokens are finite differences over these capture times. A query waits for the update of its own boundary and then runs the policy on the same GPU. While a query spans further boundaries, the tracker keeps updating at a higher GPU priority on the newest frame only, so that an update never queues behind older ones. Tracking and this contention are part of the measured latency (Table S8).

Demonstration labels. The target-state inputs of a demonstration are a causal replay of the same tracker over its recorded RGB-D at the control-period boundaries. The pose labels of the forecast are the tracked poses at the future boundaries in the same object frame, [refined offline over the episode], and the flow labels follow [3D point tracks of the recorded RGB-D, as for DROID (Appendix A.4)]. For the ball targets, the target-state tokens keep the translation, velocity, and acceleration of the center and mask the orientation and angular motion, and the pose head is supervised on translation only. Table S12 reports how often tracking was lost in the test trials.

Table S12. **Tracking on the real robot.** Per task, over the test trials of SE(3)-WAM: control-period boundaries, the fraction with an invalid estimate, the longest gap, and re-initializations.

Task	Boundaries	Invalid (%)	Longest gap (s)	Re-init.
T1	—	—	—	—
T2	—	—	—	—
T3	—	—	—	—
T4	—	—	—	—
T5	—	—	—	—
T6	—	—	—	—

D.3. Training and Evaluation Protocol

Training. SE(3)-WAM is fine-tuned per task from the DROID base (Appendix A.4). $\pi_{0.5}$, DynamicWAM, and Diffusion Policy [12] train or fine-tune on the same demonstrations with the initialization, recipe, and chunk length of Table S4, and DynamicWAM executes [with or without real-time chunking]. Demonstrations, development trials, and test trials come from separate sessions.

Table S13. **Real-robot successes and failures.** Top: successes per task (of 20 trials; tasks in Table S11); All: pooled success rate (%) with its 95% Wilson interval; Δ: paired difference from SE(3)-WAM (points) with a bootstrap interval over trial keys. Bottom: failures by category over all 120 trials of each method (Appendix D.4); –: not applicable. Shading of Δ: blue for higher SR, orange for lower, saturating at 15 points.

Method	T1	T2	T3	T4	T5	T6	All	Δ
$\pi_{0.5}$	—	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—	—
Diffusion Policy	—	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—	–

	Tracking	Late	Misaligned	Slip/drop	Collision	Timeout	Failures
$\pi_{0.5}$	–	—	—	—	—	—	—
DynamicWAM	–	—	—	—	—	—	—
Diffusion Policy	–	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—

Figure placeholder: Fig. S3 — failure cases of SE(3)-WAM

Layout: full width, two rows, one for each of the two most frequent failure categories of SE(3)-WAM in Table S13 (e.g., late, slip or drop); four equal 16:9 panels per row. **Panels per row:** the query frame, the first execution of its chunk, contact or the closest approach, and the end of the trial, each with its time after capture in ms from the trial log; overlays: blue tracked past poses and purple commanded gripper path; the task, the trial key, and the failure category above each row. **Rules:** real photos, unretouched; overlays from the logged data of the shown trials; captions state observable events only.

Figure S3. **Failure cases of SE(3)-WAM** on the real robot: [category, task, and observed event] (top) and [category, task, and observed event] (bottom).

Trials. Each task has 20 pre-generated trial keys, which fix the object instance, its initial pose, and the device setting (Table S11). The trials are blocked by key: for each key, the four methods run consecutively in a random order, so that drift of the apparatus does not align with the methods. An operator resets the scene to the key, aligned with a reference image from the camera. A trial starts at the first usable observation, the robot may move at any time, and a trial ends at success, failure, a safety stop, or the time limit, set to about three times the median duration of the task's demonstrations. A trial is repeated with the same key only if the device deviates from the key on its own, [a measured speed outside ±X% of the setting or a device fault]; everything else counts as a failure, including a tracking loss of SE(3)-WAM and a safety stop.

Statistics. We report successes per task and the pooled success rate with its 95% Wilson interval; with 20 trials per task, the pooled rate equals the task-balanced one. Paired differences from SE(3)-WAM use a bootstrap over trial keys within each task. We log the end-to-end time-to-action of every query of every method on the same RTX 5090, including tracking for SE(3)-WAM; Table 3 reports its p95 and Table S8 the timing on the robot.

D.4. Detailed Results and Failures

Table S13 reports the successes per task with intervals and paired differences, and the failures of each method by category. The categories, fixed before the test trials, are observable events: tracking lost (SE(3)-WAM only), late (the gripper closes behind the target), misaligned grasp, slip or drop after contact, collision or safety stop, and timeout. [Tasks with the largest margin and their failure categories; timing on the robot; the most frequent failures of SE(3)-WAM, Fig. S3.]

E. Design Studies

We study the history input, future supervision, and world–action interface with 4-layer models of width 768, trained for 4 epochs (40,104 updates at global batch size 16). The earlier synchronous executor executes one command per query on 1,400 development scenes, 100 per task; the initial forecast screen reports its 140-scene selection subset and the remaining 1,260 scenes separately. Encoders, inputs, and seed counts differ between studies; comparisons are within a study. The path, auxiliary, interface, and encoder

studies retain the eight-frame visual windows and GT target-motion history of their screening recipe. The final-scale, real-time ablations are reported separately in Sec. 4.3 and Appendix C.4.

E.1. History Representation

Which history helps control? With a frozen DINOv3-L encoder [35] and one training seed, we compared seven visual-history encodings against the current image alone and against the current image with the target's SE(3) history (Fig. S4a). Every condition retains the current ground-truth target pose and the robot-state history. The visual encodings include flow-based alternatives [15, 28]. The target-history condition also receives past ground-truth poses and their derived kinematics.

Target history raises SR from 53.43% to 58.86%, a gain of 5.43 points with a paired 95% scene-bootstrap interval of $[3.07, 7.86]$. None of the tested visual histories exceeds the current-image reference in point estimate. More readable history features do not necessarily improve control: for the RGB window, a frozen linear probe of the summary representation reduces the error in reading position 350 ms in the past from 10.89 to 8.46 mm, while SR falls by 6.14 points. These results support retaining explicit target motion under this training recipe.

Does the trained policy use the history? We keep the target-history checkpoint fixed, replace its past target poses by the current pose, and recompute the kinematics. Current pose, image, robot state, and timestamps remain intact. SR falls from 58.86% to 0.64% (Fig. S4b), and the first executed action changes in all 210 diagnostic windows; an identity replacement reproduces the original actions. The drop measures sensitivity to incorrect history; the training benefit is the 5.43-point comparison above. The intervals quantify scene variation conditional on these single-seed checkpoints.

E.2. Future Representation and Supervision

Initial target screen. With frozen DINOv3-L and seed 0, we compared fifteen forecast configurations (Table S14). Geometry uses eight future horizons: target SE(3), base-frame displacements of a 14×14 patch grid, or projected 2D motion, all relative to the current state. Codec targets use current-to-future frame pairs; object DINO pools features inside the target mask. Wan2.2 VAE targets temporally compress the nine-frame clip, rather than providing one latent per horizon.

On the 1,260 remaining scenes, P07 and P09 have the highest point estimates, 64.05% and 64.13%, compared with 62.78% without future supervision (P00) and 62.38% for scene flow with SE(3) (P03). The paired comparisons do not establish an overall advantage for P07 or P09. Dense 2D displacement (P05) lowers SR by 3.57 points relative to P00 (Bonferroni-adjusted $p = 0.00844$ over 27 comparisons). These are comparisons of complete forecast configurations: P07 uses direct regression from shared features, whereas P03 uses an independent Future stream within the backbone and flow matching. Their difference therefore motivates a separate comparison of representation and prediction path.

Separating representation from prediction path. We crossed 2D and 3D targets with direct regression and the Future-stream flow-matching implementation, holding the trainable CNN, SE(3) supervision, current-target RGB reconstruction, and history inputs fixed, with three training seeds per condition (Fig. S5a). Both coordinate representations use the same source points and horizons; their scales are calibrated from training labels. Direct prediction gives higher SR than the Future-stream implementation in every seed for both 2D and 3D. Within either path, the 2D–3D difference changes sign across seeds. Across paths, the loss, motion weighting, and visibility supervision also change, so the result favors the complete direct-prediction recipe.

A further comparison fixes the direct path, frozen DINOv3-L, and all three future appearance targets (Fig. S5b). The 2D–3D difference is $+2.00$, -1.50, and $+1.00$ points across seeds, with a mean of $+0.50$. We retain 3D displacement to express scene motion in the robot's metric frame; these development studies do not establish a control-performance advantage of 3D over 2D.

Selecting auxiliary supervision. With frozen MAE-B and seed 1, we keep direct 3D flow, SE(3), and current-target RGB reconstruction, and compare no future appearance target, each target alone, and all three together (Table S15). DINO and dynamic RGB alone improve on the no-auxiliary reference, depth alone lowers SR, and the combination has the highest SR, 71.07%. The single-target rows measure the effect of adding that target to the reference; they do not isolate its marginal contribution when the other two are present. This screen motivates retaining the combination with MAE-B. Auxiliary benefits depend on the encoder: in the three-seed CNN study, all three targets average 62.86%, versus 63.95% with none. The final-model no-future control in Sec. 4.3 tests the full forecast supervision set, including the geometric targets.

E.3. World–Action Information Flow

Interfaces and controls. With frozen MAE-B and shared attention with separate world/action feed-forward networks, we distinguish observed conditions C, future-query hidden states W, and decoded predictions Y. Actions always read C. One object query and one query per patch read C and one another to predict SE(3) and scene flow. R1–R4 keep the modules, losses, and initialization procedure fixed and toggle the two attention edges $W \to A$ and $A \to W$ (Table S16).

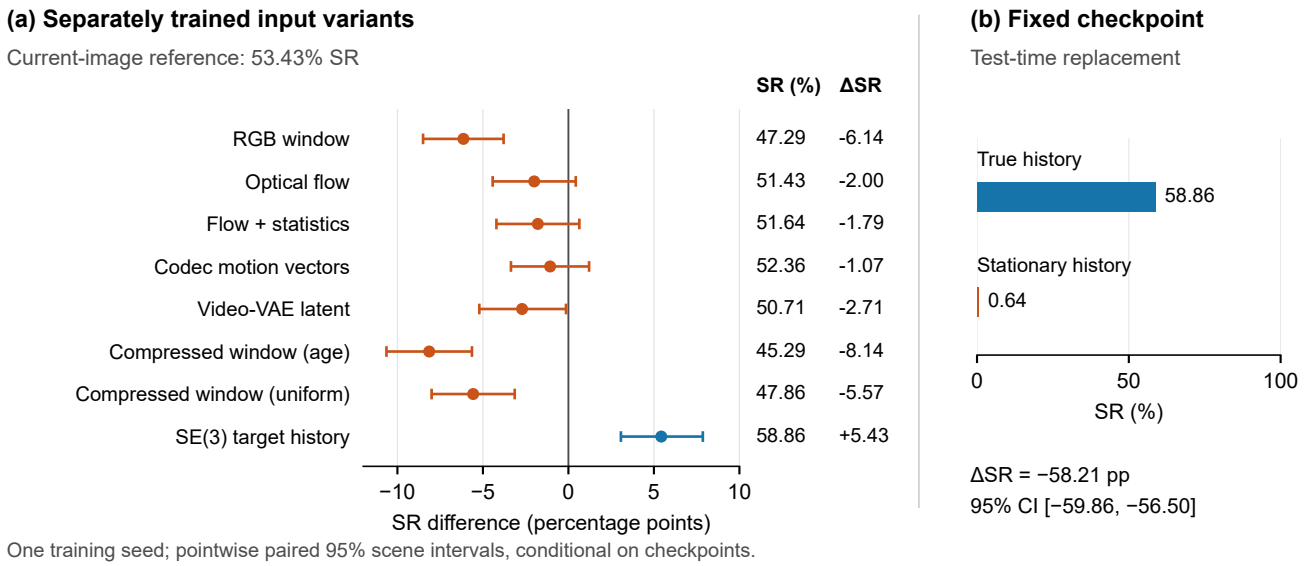


Figure S4. **History input and test-time intervention.** Frozen DINOv3-L, one training seed, 1,400 development scenes. All conditions retain the current GT target pose and robot history. (a) Separately trained variants: SR and its difference from the current-image reference. Whiskers are pointwise paired 95% scene-bootstrap intervals, conditional on trained checkpoints, without multiplicity adjustment. (b) Replacing history at the same checkpoint: 824 versus 9 successes out of 1,400. This measures sensitivity to incorrect history, separately from the training benefit in (a).

Table S14. **Initial forecast-configuration screen.** Frozen DINOv3-L, seed 0. The 140-scene selection subset and the remaining 1,260 development scenes are disjoint. Rows are grouped by the changed target; prediction paths also differ, as described in Appendix E.2. A dash means that target is absent. Blue/bold and light blue mark the best and second point estimates within each outcome column, not statistical significance.

ID	Scene target	Object target	Additional target	Success / 140	SR_{1260} (%)
Basic supervision controls					
P00	–	–	–	88	62.78
P01	3D scene flow	–	–	**95**	62.06
P02	–	SE(3)	–	92	62.62
P03	3D scene flow	SE(3)	–	93	62.38
Scene-target alternatives to P03					
P04	Video-VAE latent	SE(3)	–	92	62.14
P05	Dense 2D displacement	SE(3)	–	87	59.21
P06	Codec motion vectors	SE(3)	–	86	63.10
P07	Sparse 2D tracks	SE(3)	–	90	64.05
Object-target alternatives to P03					
P08	3D scene flow	Object DINO	–	89	63.02
P09	3D scene flow	Target 2D tracks	–	88	**64.13**
Additional targets on P03					
A_α	3D scene flow	SE(3)	Motion magnitude	89	61.75
P12	3D scene flow	SE(3)	Depth	90	62.78
P13	3D scene flow	SE(3)	DINO	92	62.54
P14	3D scene flow	SE(3)	Dynamic RGB	89	62.70
P15	3D scene flow	SE(3)	Depth, DINO, dynamic RGB	94	63.65

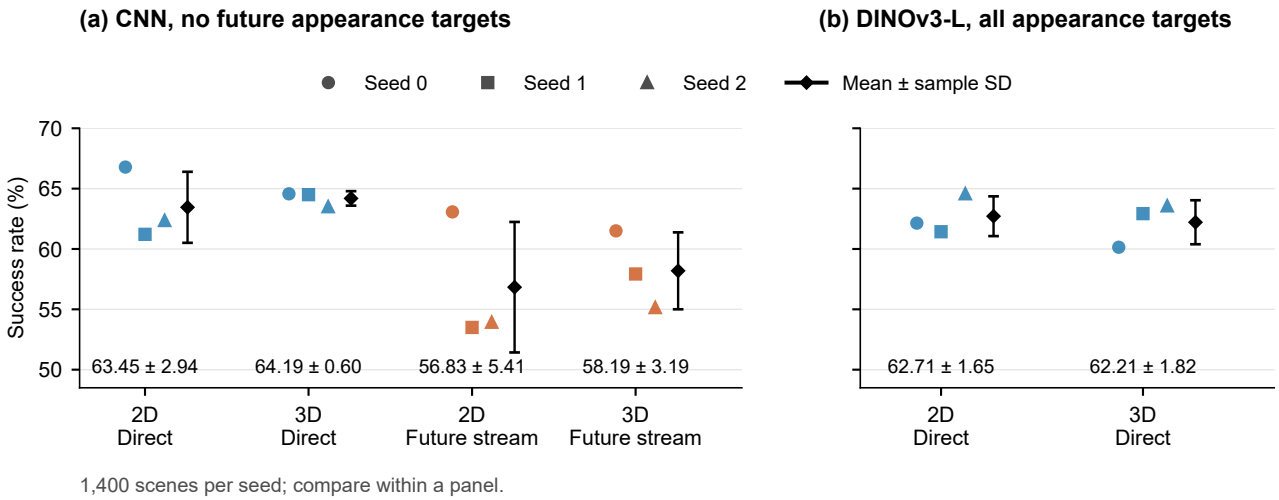


Figure S5. **Representation and prediction path.** Colored markers show three training seeds, each evaluated on 1,400 development scenes; black diamonds and whiskers show their mean and sample standard deviation, also printed below. Both panels retain SE(3) and current-target RGB supervision. (a) CNN without future appearance targets: direct regression versus the Future-stream flow-matching recipe. (b) Frozen DINOv3-L, direct regression, and all three appearance targets. Compare within each panel; the seed spread is not a confidence interval.

Table S15. **Future appearance supervision.** Frozen MAE-B, seed 1, 1,400 scenes; direct 3D flow, SE(3), and current-target RGB reconstruction stay enabled. Δ: SR difference from the first row (points). ✓: enabled; –: disabled. Blue/bold and light blue mark the highest and second SR point estimates. Δ shading: blue for higher SR, orange for lower, saturating at 15 points; it does not encode significance.

Depth	DINO	Dynamic RGB	SR (%)	Δ
–	–	–	63.86	ref.
✓	–	–	61.93	−1.93
–	✓	–	67.57	+3.71
–	–	✓	66.14	+2.29
✓	✓	✓	**71.07**	+7.21

Table S16. **World–action information flow.** Frozen MAE-B, three training seeds, 1,400 development scenes per seed. W: future-query hidden states; A: action states; Y: decoded geometry. Actions always read observed conditions C. R1–R4 form the two-edge comparison. SR is the seed mean ± sample s.d. (%); Δ_{R1} is the mean ± sample s.d. of same-seed differences from R1 (points). Blue/bold and light blue mark the highest and second SR point estimates. Δ shading: blue for higher SR, orange for lower, saturating at 15 points; it does not encode significance.

ID	Future losses	$W \to A$	$A \to W$	$Y \to A$	SR (%)	Δ_{R1}
R0	–	–	–	–	60.45 ± 3.57	-3.14 ± 5.75
R1	✓	–	–	–	63.60 ± 2.34	ref.
R2	✓	✓	–	–	62.52 ± 1.97	-1.07 ± 4.28
R3	✓	–	✓	–	$\mathbf{64.21 \pm 2.13}$	$+0.62 \pm 0.84$
R4	✓	✓	✓	–	63.00 ± 4.19	-0.60 ± 6.52
R5	✓	–	–	✓	63.95 ± 1.91	$+0.36 \pm 3.45$

Their recorded joint evaluation recomputes W at each Euler step. R0 keeps the same modules but disables all future losses; current RGB reconstruction remains. R5 instead decodes Y once per query and re-encodes the trajectory, flow, and visibility into action conditions with a 0.37M-parameter geometry encoder. It uses predicted geometry during both training and inference, with action gradients flowing through that interface.

Supervision and action conditioning. R1 improves on R0 by 3.14 points on average, with seed-wise differences of −1.29, +9.64, and +1.07. Adding hidden-state access (R2) or decoded geometry (R5) changes SR relative to R1 by −1.07 and +0.36 points, respectively, with seed-wise difference standard deviations of 4.28 and 3.45 points. Neither produces a consistent gain across seeds. R3 has the highest mean, but its 0.62-point gain over R1 becomes −0.03 points outside the phase-only wire task; all fourteen tasks remain in the primary metric. The main effects of the two edges and their interaction also change sign across seeds.

Relation to the deployed model. We retain future supervision without a forecast-to-action interface. R1 uses independent future queries; SE(3)-WAM reads forecasts directly from the existing scene and summary tokens (Sec. 3.3). These distinct architectures share the supervision-only use of forecasts. In SE(3)-WAM, actions still read the observed world representation, while forecast heads are skipped at

deployment.

E.4. Encoder and Parameter Sharing

Table S17. **Encoder and parameter sharing.** Three seeds, 1,400 scenes per seed; direct 3D, SE(3), and all appearance targets. SR: seed mean ± sample s.d. (%). Panel (b) fixes MAE-B, reusing (a)'s shared-attention runs. Params: trainable backbone; ⋆: selected. Blue/bold and light blue mark the best and second point estimates within each panel and outcome column, not statistical significance.

Configuration	Status / params	SR (%)
(a) Observation encoder		
CNN	trainable	62.86 ± 2.64
DINOv3 ViT-S/16	frozen	61.69 ± 2.00
DINOv3 ViT-L/16	frozen	62.21 ± 1.82
MAE ViT-B/16⋆	frozen	$\mathbf{66.76 \pm 3.84}$
(b) World/action parameter sharing		
Shared attention⋆	47.266M	$\mathbf{66.76 \pm 3.84}$
Separate parameters	47.272M	62.86 ± 2.98

Observation encoder. We compare a trainable CNN with frozen DINOv3-S, DINOv3-L [35], and MAE-B [17], keeping direct 3D prediction, SE(3), and the three appearance targets (Table S17a). Each uses three seeds and the same 8-GPU training recipe and global batch size. MAE-B has the highest mean SR, 66.76%, and is selected for the subsequent interface and backbone studies.

Parameter sharing. With MAE-B fixed, we compare shared attention and stream-specific feed-forward networks against separate world/action attention and feed-forward parameters (Table S17b). Both keep the same information flow and about 47.3M trainable backbone parameters; the separate variant uses a smaller feed-forward width to match this count. Shared attention averages 66.76% SR, against 62.86% for separate parameters. Their mean gap falls to 0.39 points when both wire tasks are excluded, so the difference is concentrated in those tasks. We retain shared attention based on this aggregate comparison.